

PCB/CCL Update: Will High-speed Copper Foil Become Another Bottleneck?

We estimate HVLP demand to expand from ~1.3kt/mo in 2025 to >6kt/mo in 2030, driven by 30-40% CAGR of AI CCL demand for infra buildout. Spec upgrades from HVLP1/2/3 to HVLP4 and wider adoption of DTH are key drivers of soaring ASP. Industrywide supply tightness to create opportunities for Chinese vendors on high-end copper foil to fill the gap for 3.5-4.0kt/month of HVLP4 equivalent demand in 2027. Shengyi is a major beneficiary in our coverage.

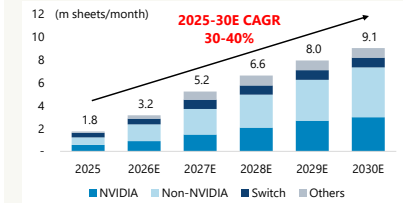
Copper foil to dance with AI PCB/CCL. Echoing our previous initiation on PCB ([Link](#)), given we forecast ~60%/70% CAGR of AI PCB/CCL TAM in 2025-30E driven by tech giants' rising capex plans on AI infra buildout, this trend has also brought structural changes to electrolytic copper foil as one of key upstream materials. We estimate AI CCL monthly demand at ~1.8m sheets/month in 2025 and expect 30-40% CAGR to reach >9m sheets/month in 2030. As a result, we forecast demand of HVLP, as a critical high-speed copper foil for AI infra with significantly higher value and technical barrier vs traditional copper foil, to grow from ~1.3kt/month in 2025 to >6kt/month in 2030.

Soaring ASP from spec upgrade a key driver. Major AI players (NVIDIA/Google/Amazon) will largely transfer their copper foil spec from HVLP1/2/3 to HVLP4 on their upcoming new AI servers in 2H26, creating significant demand on the latter. The outlook from Mitsui on its HVLP sales growth CAGR at 15-20% in 2025-30 seems conservative, potentially creating spillover opportunities for other vendors. We also see rising demand on DTH as another advanced copper foil driven by memory and optical transceiver. We estimate HVLP4 could reach >10x as of traditional HTE on pricing, and DTH could even be roughly 2x as of HVLP4. As demand continues to shift toward higher-spec HVLP/DTH going forward, driven by AI infra construction, we believe the copper foil industry's ASP will continue to rise.

AI copper foil supply gap could be there, but Chinese vendors are gaining traction to fill it up. With industrywide supply tightness caused by surging AI-related demand, we think domestic players could demonstrate their advantage in production capacity and potentially enable them to gain more traction on high-end copper foil. We anticipate notable supply gap in the next 1-2 years for HVLP4-equivalent demand if we only count international players' capacity. But by including TGCF/Defu as emerging suppliers on high-speed copper foil, we now expect global major players' HVLP capacity to increase from ~2kt/month in 2025 to >5kt/month in 2027 (or 3.5-4.0kt/month of HVLP4-equivalent capacity), roughly on a par with demand by then, though it still depends on each vendor's execution on HVLP4's capacity ramp.

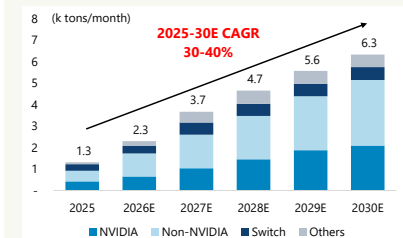
Implications on the industry & our coverage. We believe such trend will benefit those copper foil and equipment vendors technically redied. **TGCF/Defu** are now leading Chinese players in AI copper foil in customer verification and shipment volume. For TGCF, its RTF/HVLP already took up the majority of its gross profit in 2025, making it almost a direct AI play. For Defu, in addition to capabilities for high-speed copper foil, we see an active capacity expansion plan (another 50kt capacity in 2027-28 focusing on RTF/HVLP), which may demonstrate its strength in acquiring critical tools. Equipment makers such as **Taijin New Energy** could also have potential spillover opportunities. Among our coverage, we expect **Sytech** to have advantages in securing sufficient materials going forward as it continues to diversify its supply sources (Chinese upstream suppliers are more proactive in expanding capacity vs international vendors), enabling it to further expand its market share in AI CCL supply chain.

Chart 1 - Global AI CCL Demand Forecast



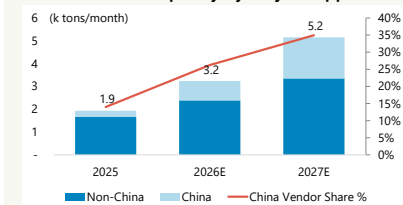
Source: Jefferies estimates

Chart 2 - Global AI HVLP Demand Forecast



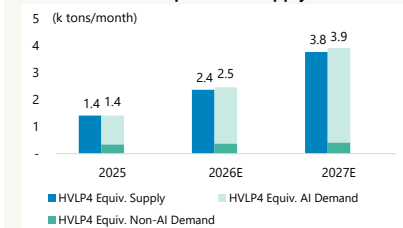
Source: Jefferies estimates

Chart 3 - HVLP Capacity by Major Supplier



Source: Company data, Jefferies estimates

Chart 4 - HVLP4 Equivalent Supply vs Demand



Source: Company data, Jefferies estimates

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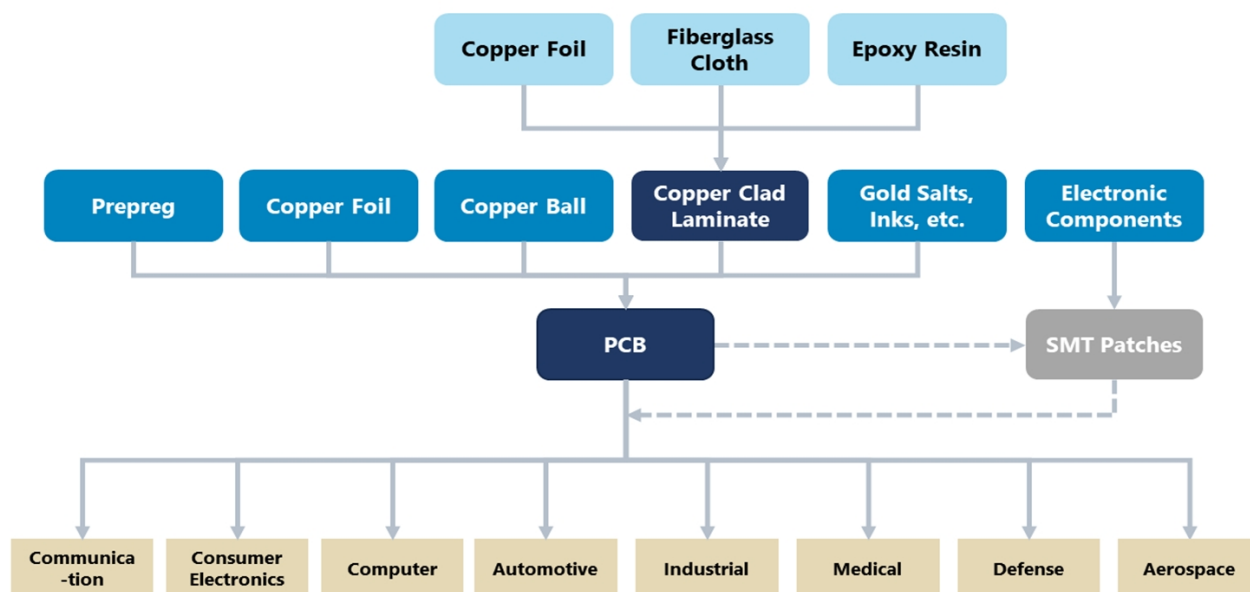
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Copper Foil's Revival: Riding the Tide of Booming AI PCB Demand

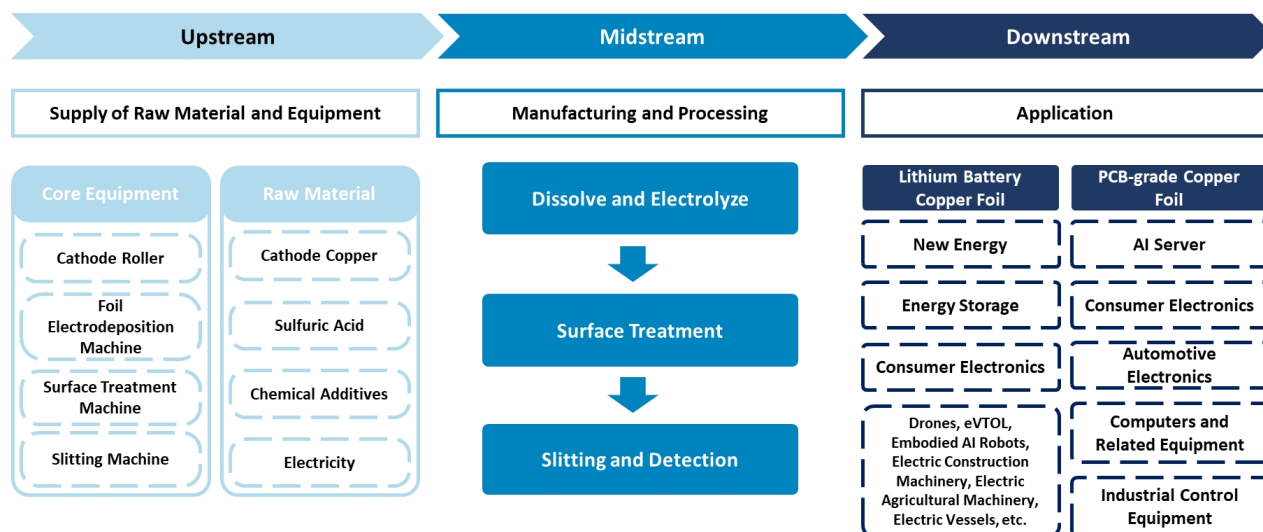
Copper foil to ride on booming AI PCB/CCL demand. As indicated in our initiation on several PCB names ([Link](#)), we forecast ~15% CAGR in 2025-30 for the global TAM of PCB, mainly driven by ~60%/70% CAGR of AI PCB/CCL TAM, as AI infrastructure growth is likely to continue to be supported by tech giants' rising capex plans. We believe this trend has also brought structural changes to electrolytic copper foil, which acts as one of key upstream materials for PCB/CCL to be adopted on computing infra (represented by data center equipment such as server, switch, optical transceiver, etc.). In addition to PCB, copper foil is also widely adopted in LIB (lithium battery), and we observe the industry demand is accelerating as well, further fueling the expansion of copper foil's market size.

Chart 5 - PCB Supply Chain



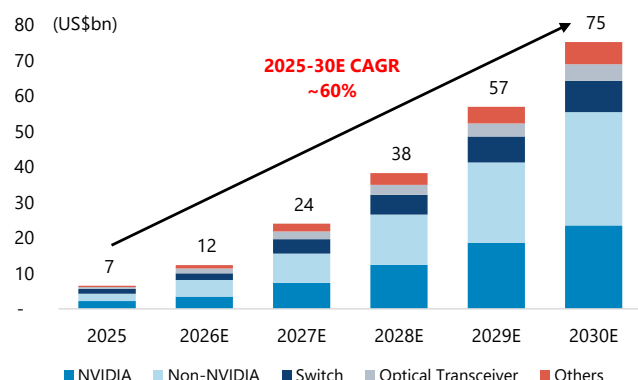
Source: Company data, Jefferies

Chart 6 - Electrolytic Copper Foil Supply Chain



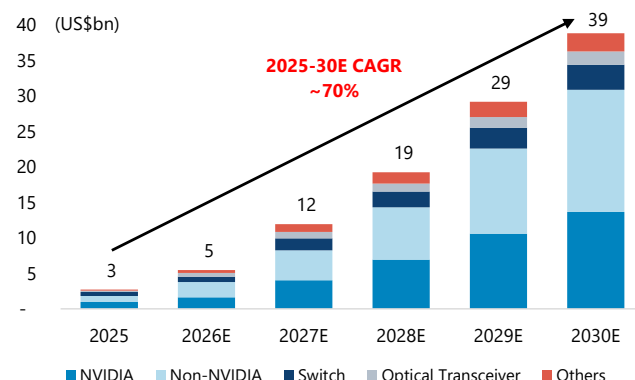
Source: Company data, Jefferies

Chart 7 - Global AI PCB TAM Forecast



Source: Prismark, Jefferies estimates

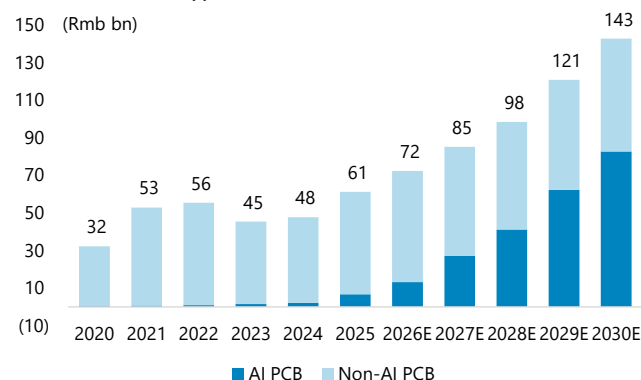
Chart 8 - Global AI CCL TAM Forecast



Source: Prismark, Jefferies estimates

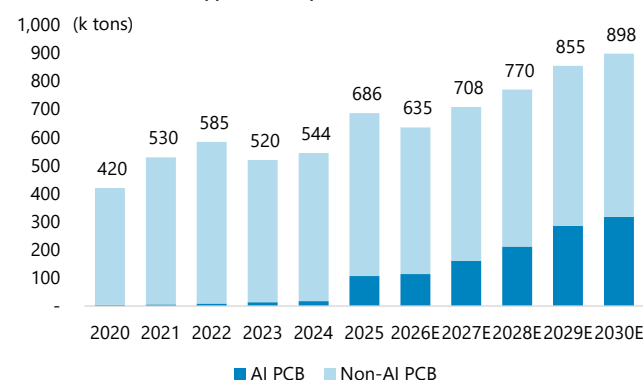
We forecast CAGR of 30-40% for AI HVLP shipment. To meet the rising requirement of data transmission, HVLP (high-volume low-pressure) copper foil as a high-end material has already become the mainstream solution for AI infra, which has significantly higher value and technical barriers compared with traditional copper foil, and we see continued product upgrade for stronger performance going forward. Based on current AI chip volume and allocation to each player (NVIDIA, CSP, etc.), AI PCB spec adopted and future roadmap, we estimate AI CCL monthly demand at **~1.8m sheets/month** in 2025 (including GPU/ASIC server, high-speed, and others), translating to **~1.3k tons/month** demand on HVLP. We then forecast **30-40% CAGR** for HVLP's demand to reach **>6k tons/month** in 2030, assuming 1) AI players continue to afford rising capex; 2) ongoing spec upgrade for increasing dollar content; and 3) smooth capacity expansion by major copper foil vendors without significant disruption in supply chain.

Chart 9 - Global PCB Copper Foil TAM Forecast



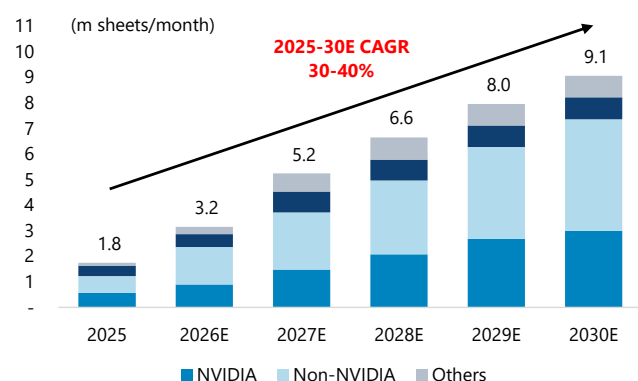
Source: CCFA, Jefferies estimates

Chart 10 - Global PCB Copper Foil Shipment Forecast



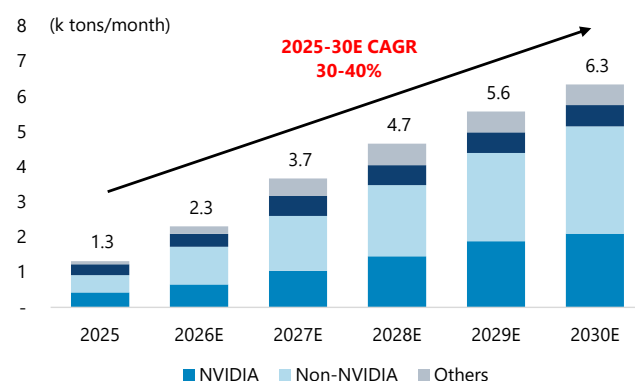
Source: CCFA, Jefferies estimates

Chart 11 - Global AI CCL Demand Forecast



Source: Jefferies estimates

Chart 12 - Global AI HVLP Demand Forecast



Source: Jefferies estimates

Moving towards Higher-end: From HTE/RTF to HVLP/DTH

Materials upgrade to push PCB/CCL's evolution. The major end devices that adopt high-end PCB include AI server (mainboard, OAM, UBB, etc.), high-speed switch, and optical transceiver ($\geq 400\text{G}$), and copper foil generally takes up $\sim 30\%$ of CCL's cost and another $< 5\%$ of PCB's cost (adopted on both PCB/CCL individually). While HTE (high-temperature elongation) acts as a commodity-like product with wide applications in many downstream markets and RTF (reverse-treated foil) used to be a major solution on general server/switch, high-end materials such as HVLP and DTH have now been widely adopted in AI infra, thanks to its much stronger signal transmission capabilities, though at a higher cost, enabling better performance of PCB/CCL on AI server/switch, etc.

Table 1 - PCB Copper Foil Spec Comparison

Parameters	HTE	RTF	HVLP
Roughness	$> 4\mu\text{m}$ (Rz)	Smooth surface 0.5-4 μm (Rz)	$\leq 1.0\mu\text{m}$ (Rz)
Core advantage	High temperature elongation	Simplified manufacturing process	Ultra-low signal loss
Applicable frequency	Low to mid frequency	Mid to high frequency	High to ultra-high frequency
Main defect	High frequency loss	Signal loss higher than HVLP	Poor lamination adhesion
Cost	Low	Medium	High

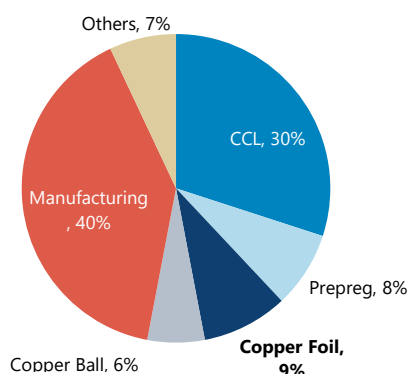
Source: GPCA, Jefferies

Table 2 - CCL Spec By Grade

Grade	Ethernet	Loss	Dk	Df	Fiberglass	Copper Foil	Application
FR4	$< 5\text{Gbps}$	Standard Loss	~ 4.7	0.023-0.015	E Glass	RTF	General
M2	8Gbps	Medium Loss	~ 4.3	0.015-0.010	E Glass	RTF	PCIe 2.0, 10G Ethernet
M4	10Gbps	Low Loss	~ 3.9	0.010-0.005	E Glass	RTF/HVLP1	PCIe 3.0, 10G/100G Ethernet
M6	25Gbps	Very Low Loss	~ 3.8	0.005-0.003	E Glass	HVLP2	PCIe 5.0, 100G/400G Ethernet
M7	56Gbps	Ultra Low Loss	~ 3.5	0.003-0.0015	Low-Dk1	HVLP2/3	PCIe 6.0, 400G/800G Ethernet
M8	112Gbps	Extreme Low Loss	~ 3.0	0.0015-0.0010	Low-Dk1/2	HVLP2/3/4	PCIe 7.0, 800G/1.6T Ethernet
M9	224Gbps	/	< 3.0	0.0009-0.0007	Q-Glass/Low-Dk2	HVLP4/5	$> \text{PCIe 7.0}, > = 1.6\text{T Ethernet}$

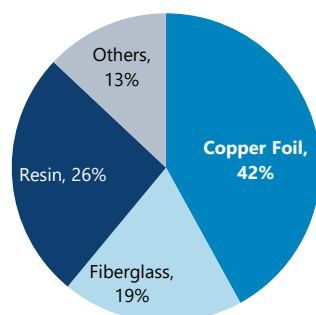
Source: Company data, Jefferies

Chart 13 - Conventional PCB Cost Structure



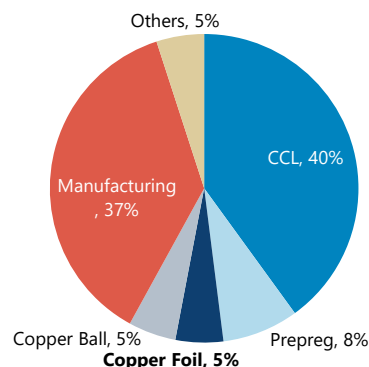
Source: AskCI Consulting, Jefferies

Chart 15 - Conventional CCL Cost Structure



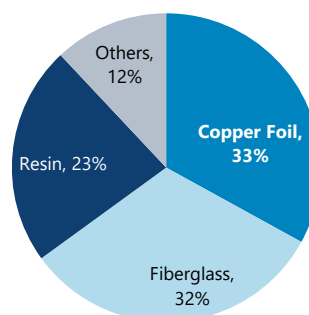
Source: AskCI Consulting, Jefferies

Chart 14 - A Certain High-end PCB Cost Structure



Source: Jefferies estimates

Chart 16 - A Certain High-end CCL Cost Structure



Source: Jefferies estimates

HVLP to migrate from gen 1 to gen 4; International leader's outlook seems conservative. As a typical high-end copper foil, HVLP is generally differentiated by the surface profile (including adhesion, roughness, etc.) as a key property. The higher the spec, the lower the transmission loss

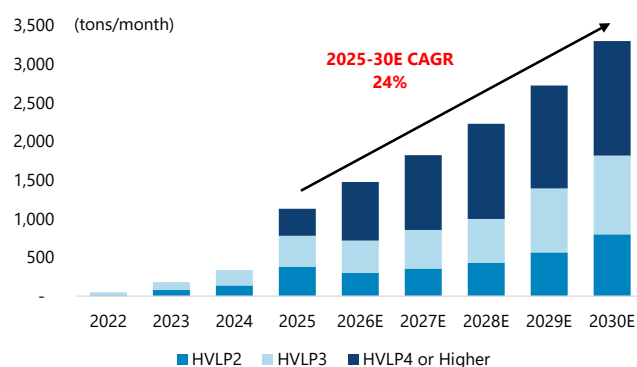
will be for HVLP. After years of product evolution, HVLP has been upgraded from the initial gen 1 to gen 4, which has now been gradually deployed in mass volume (HVLP5 also in R&D stage in the industry, poised for commercialization likely in next 1-2 years). Our channel checks suggest major AI players, including NVIDIA, Google, and Amazon, will largely transfer their copper foil spec on compute/switch boards from originally HVLP1/2/3 to HVLP4 on their upcoming new AI servers in 2H26, creating significant demand on the latter going forward. Mitsui Kinzoku is currently the largest supplier of HVLP with ~60% market share in **HVLP2 and above** in 2025 and running at ~600t/month of HVLP capacity, per the company, while its outlook on its HVLP sales growth CAGR at 15-20% in 2025-30 seems conservative compared with our forecast of 30-40% demand CAGR, potentially creating spillover opportunities to other vendors for market share expansion, in our view.

Table 3 - Major AI Server Copper Foil Spec/Supplier

Product	Content	CCL Spec	Fiberglass Spec	Copper Foil Spec
NVIDIA				
GB200/300	Compute Board	M8	Low-Dk1	HVLP3
	Switch Board	M8	Low-Dk2	HVLP2
VR200	Compute Board	M8	Low-Dk1	HVLP4
	Switch Board	M8.5	Low-Dk2	HVLP4
	Midplane	M8.5	Low-Dk2	HVLP4
VR300	Backplane	M9/10/PTFE?	Q Glass/Low-Dk2?	HVLP4/5?
	CoWoP	M8/9/10?	Q Glass/Low-Dk2?	HVLP4/5?
Google				
v5	UBB	M7	Low-Dk1	HVLP2
v6/v7	UBB	M7	Low-Dk1	HVLP2
v8	UBB	M8	Low-Dk1/2?	HVLP4
v9	OAM	M9/10/PTFE?	Q Glass/Low-Dk2?	HVLP4/5?
Amazon				
Trainium 2	OAM	M6	Low-Dk1	HVLP2
	UBB	M8	Low-Dk1	HVLP2
Trainium 2.5	OAM	M6	Low-Dk2	HVLP3
	UBB	M8	Low-Dk2	HVLP4
Trainium 3	OAM	M7	Low-Dk2	HVLP4
	UBB	M8	Low-Dk2	HVLP4

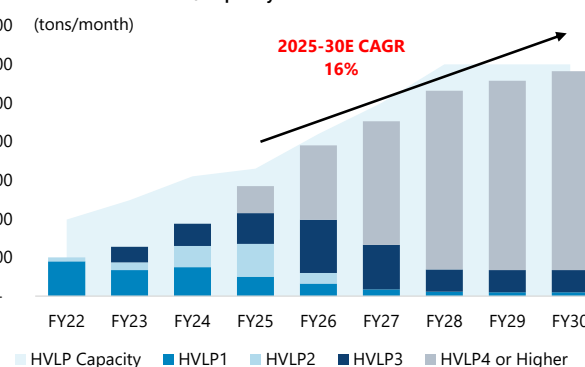
Source: Company data, Jefferies estimates

Chart 17 - Mitsui HVLP Market Demand Forecast



Source: Company data, Jefferies

Chart 18 - Mitsui HVLP Sales/Capacity Plan



Source: Company data, Jefferies

DTH as another higher-end product facing surging demand. In addition to HVLP, we see rising demand on DTH (double thin, or carrier-supported), which is an ultra-thin functional copper layer (typically 1.5-5um) supported by a thicker carrier foil. DTH is mainly used in advanced PCB and IC packaging to enable ultra-fine-line trace manufacturing, with concentrated adoptions on smartphone's substrate (SoC, memory, RF, etc.) and mainboard (SLP). However, we also see increasing demand from AI infra, especially on memory and high-speed optical transceiver (800G/1.6T). Currently, Mitsui has >95% market share on **DTH**, and we expect potential new

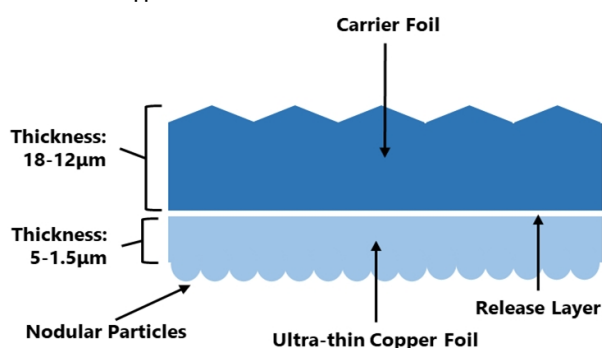
entrants to be introduced going forward, due to supply tightness, though the technical barrier is considerably high.

Table 4 - DTH Copper Foil Application

Equipment	Substrate	Target L/S (um)		
		>30/30um	30/30 to 10/10 um	10/10 um>
Smartphone	SoC	-	○	○
	DRAM (LPDDR)	-	○	-
	Mobile NAND	-	○	-
	RF Module	-	○	-
	mmW Antenna and AiP	-	○	-
Server/PC	CPU	-	-	-
	GPU	-	-	-
	DRAM (DDR)-DIMM	-	○	-
	NAND-SSD	-	○	-
	Controller-SSD	-	○	-

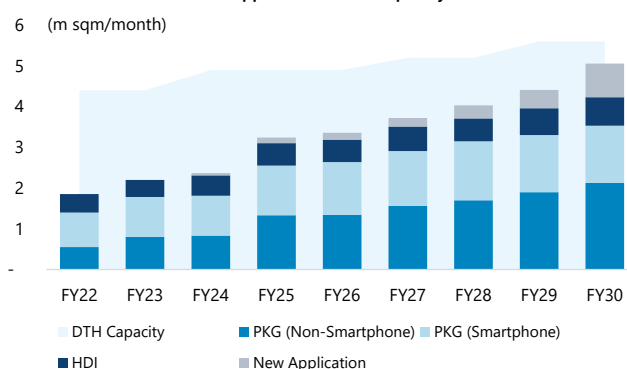
Source: Company data, Jefferies

Chart 19 - DTH Copper Foil Structure



Source: Company data, Jefferies

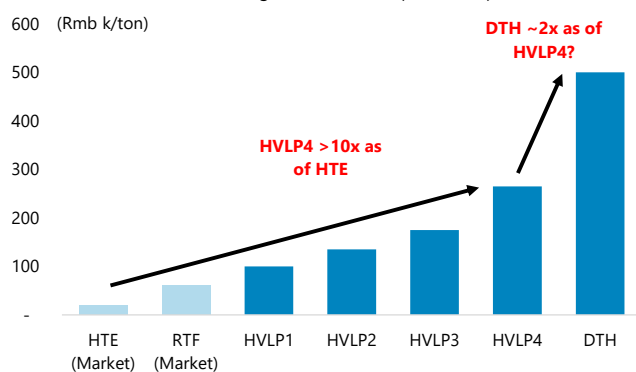
Chart 20 - Mitsui Ultra Thin Copper Foil Sales/Capacity Plan



Source: Company data, Jefferies

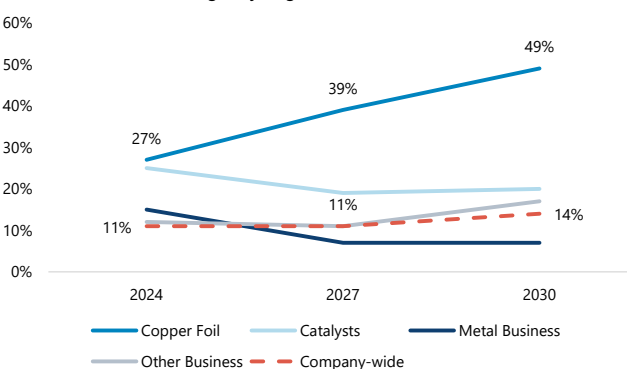
Significantly higher ASP from HVLP/DTH vs traditional products. The copper foil's pricing structure is made up by copper's price (fully transferred to the downstream) and processing fee. Given the considerably higher technical barrier on HVLP/DTH (such as surface treatment technique) and lower throughput efficiency (as a result of lower line speed, lower yield rate, etc.) compared with traditional products (HTE/RTF), there could be significant processing fee premium from the former. Due to gradually tightening supply, we see ongoing price hikes on most copper foil products in general in the industry from 2025. As of 1Q26, we estimate the market's processing fee of HTE/RTF was roughly **Rmb10-20k/30-80k/t** (depending on spec), while **HVLP4** could reach **>10x** as of HTE based on Mitsui's pricing. While DTH's processing fee is generally counted in area (square meter), we estimate **DTH** could be roughly **2x** as of HVLP4 by translating to the weight for comparison. As demand continues to shift toward higher-spec HVLP/DTH going forward, driven by AI infra construction, we believe the copper foil industry's ASP will continue to rise; Mitsui's outlook on rising ROIC of copper foil segment echoes this trend.

Chart 21 - HVLP/DTH Processing Fee Estimates (End-1Q26)



Source: Company data, Jefferies estimates

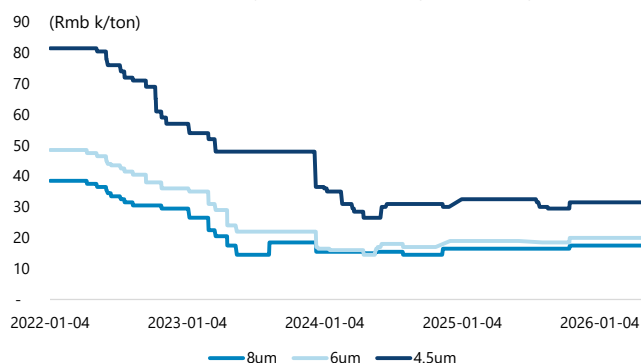
Chart 22 - Mitsui ROIC Target by Segment



Source: Company data, Jefferies

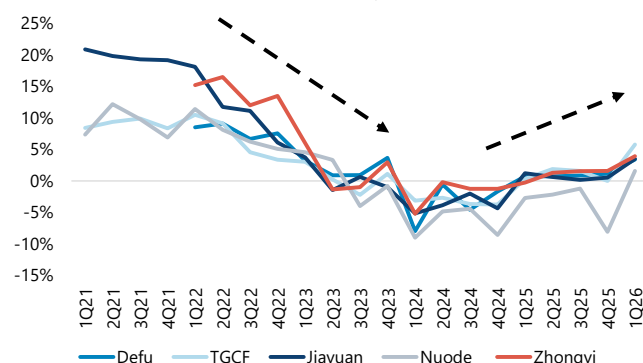
Overall copper foil bottoming out with accelerated demand growth and disciplined capacity expansion. LIB is the other key downstream market for copper coil (10-15% of total cost in general). Due to aggressive capacity expansion from multiple LIB copper foil suppliers in earlier years amid rapid penetration of EV/ESS, the entire industry had been in a state of oversupply since 2022, reflected in the slump in processing fees and utilization rate. With many players are subsequently operating at a marginal profit or even loss, capacity expansion has been gradually controlled in the industry. As a result, we think the profitability of most suppliers has rebounded since 2024. Driven by the gradual recovery of downstream demand with strong performance in some subsectors, such as energy storage, we observe the utilization rates of many copper foil suppliers are returning to higher levels, even with potential signal of price hike going forward.

Chart 23 - China Lithium Battery Copper Foil Average Processing Fee Trend



Source: Wind, Jefferies

Chart 24 - China Copper Foil Vendor Net Margin Trend

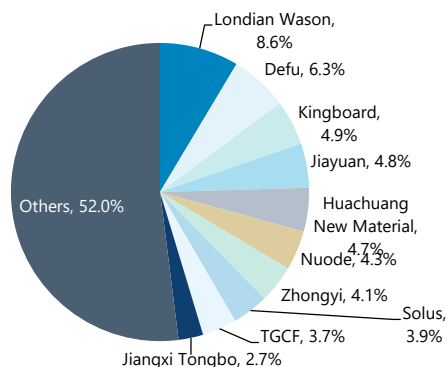


Source: Company data, Jefferies

Supply Tightness Going on: Chinese Players Are Catching Up on High-end

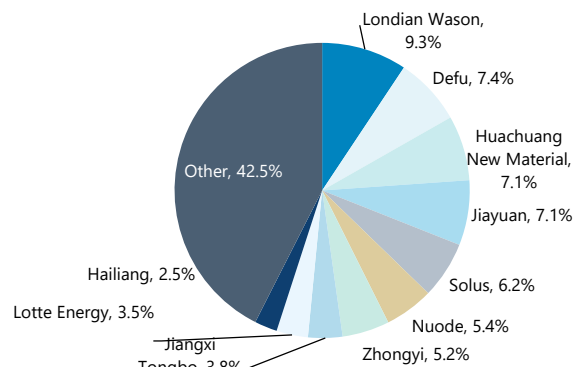
Chinese players lead in overall copper foil capacity, but with low exposure to mid-/high-end products. In terms of overall copper foil shipment, Chinese suppliers hold the majority share (9 out of the top 10 global players in 2024), with a focus mainly on the LIB market. However, the mid-/high-end copper foil market including RTF/HVLP is still dominated by suppliers outside China Mainland, represented by Japan, Korea, and Taiwan. Previously, due to the limited downstream applications of mid-/high-end copper foil, Chinese vendors had few opportunities in this field. However, thanks to the surging demand for AI-related mid-/high-end copper foil, we see domestic players could also demonstrate their advantage in production capacity and potentially enable them to gain more tractions to be introduced to AI customers' supply chains.

Chart 25 - Global Copper Foil Shipment Market Share by Vendor (2024)



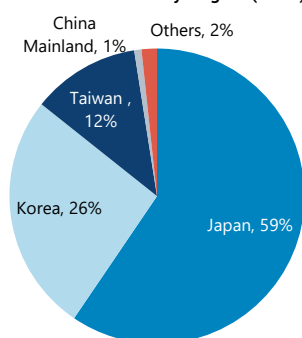
Source: Company data, Jefferies

Chart 26 - LIB Copper Foil Shipment Market Share by Vendor (2024)



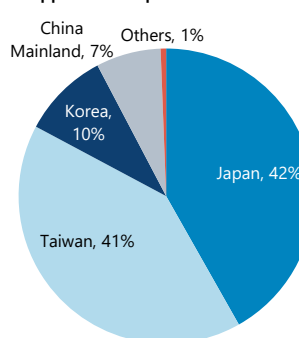
Source: Company data, Jefferies

Chart 27 - HVLP Shipment Market Share by Region (2022)



Source: Jefferies

Chart 28 - High-speed Copper Foil Shipment Market Share by Region (2022)



Source: CPCA, Jefferies

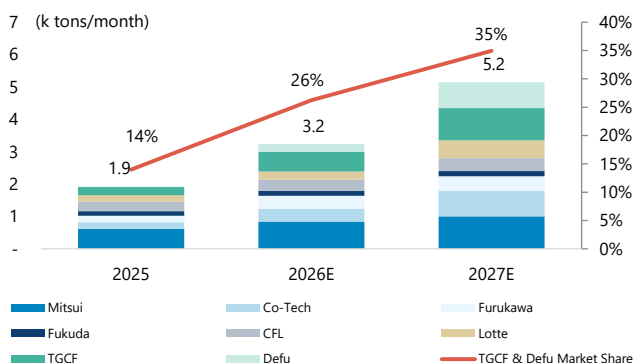
Table 5 - High Speed Copper Foil Production Line

Type	Roughness (um)	Mitsui	Furukawa	Fukuda	CFL	Lotte	Co-Tech	Chang Chun	Defu	TGCF
RTF1	<= 5.0	MLS-G	DGTSEU2-MP	TGFB-DSTF	TZA-B		RT311	RTF-3	RHS1	✓
RTF2	<= 2.3	MLS-G3		T9-DS2	TZA-B2		RG311	RTF-25	RHS2	✓
RTF3	<= 2.1	MLS-G4		T9-DS3	TZA-B3	HRT	RG312	RTF-35	RHS3	✓
RTF4	<= 1.7	MLS-G5		T9-DS4		MRT	RG313	RTF-45	RHS4	✓
RTF5	<= 1.5					SRT	RG315	RTF-55		✓
HVLP1	<= 2.0	HS1-VP	FV-WS	T9FB-SV	BF-TZA	LH1	VL410	VFP-111	VHS1	✓
HVLP2	<= 1.5	HS2-VSP	FZ-WS	T9S-SV	BF-ANP	LH2	VG410	VFP-101	VHS2	✓
HVLP3	<= 1.0	SI-VSP		DSV-A	BFL-NN	LH3	VL411	VFP-200	VHS3	✓
HVLP4	<= 0.8	SI2-VSP		T9DA-SV	BFL-NX	LH4	PF511	VFP-250	VHS4	✓
HVLP5	<= 0.7	SI3-VSP		T9DC-SV	BFL-NF	LH5				

Source: Company data, Jefferies

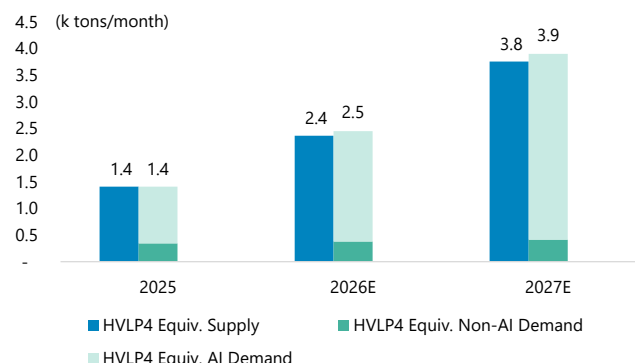
Will there be supply gap for AI copper foil? Chinese vendors are filling it up (though not 100% guarantee depending on execution). To meet the strong demand for HVLP, we see major global suppliers have developed capacity expansion plans for the next few years, either through brownfield construction or conversion from existing traditional production lines. However, by compiling capacity expansion plans from **major international HVLP vendors only**, considering the certain capacity reduction caused by spec upgrades, we anticipate the HVLP4 equivalent demand (as the next-gen mainstream solution) will still see a **notable supply gap** in the next 1-2 years. Thus, we witness a new trend where AI players are actively bringing in new suppliers, including certain Chinese players that are technically capable. Specifically, **TGCF** (Tongguan Copper Foil) entered AI supply chain previously by supplying mainly to CCL vendors such as Sytech/EMC/TUC, and **Defu** has also gradually passed the verifications of multiple PCB/CCL customers and begun to penetrate the AI customers. By including the capacity plans of TGCF/Defu, we now expect the global major players' **HVLP** capacity to increase from **~2kt/month in 2025 to >5kt/month in 2027**. This could roughly translate to **3.5-4.0kt/month of HVLP4-equivalent** capacity, roughly on par with the demand by then. However, this assumption rests on 1) each supplier's capacity expansion meeting its original expectations and 2) each supplier having the ability to convert to HVLP4 capacity with sufficient yield/efficiency. Due to current supply shortage of critical tools in the industry, especially surface treatment machine dominated by Japan's Mifune Corp (>70% market share), with limited capacity and the high technical challenge of manufacturing HVLP4, we think a supply gap is still likely to emerge in the future. Therefore, the execution of each supplier is worth monitoring.

Chart 29 - HVLP Capacity by Major Supplier



Source: Company data, Jefferies estimates

Chart 30 - HVLP4 Equivalent Supply vs Demand

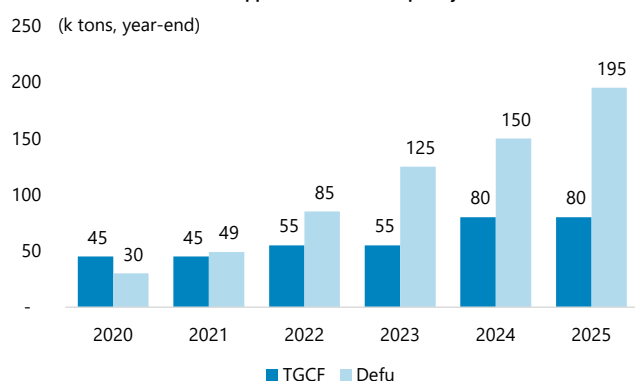


Source: Company data, Jefferies estimates

Supply chain opportunities: TGCF currently leading in AI among domestic suppliers while Defu has more capacity advantages; rising demand on critical tools to benefit local vendors like Taijin.

Based on public information, TGCF and Defu are now leading Chinese players in AI copper foil in terms of customer verification and shipment volume. For TGCF, its high-speed copper foil (RTF/HVLP) took up >20% of its total copper foil shipment in 2025 and the majority of total gross profit, making it the player with the highest exposure to high-speed copper foil among domestic vendors, and it has stable partnerships with other AI supply chain players based on its disclosure. On the other hand, Defu, although it is a relative newcomer in this area, already possesses competitive production capabilities for high-speed copper foil with an active capacity expansion plan. Although both companies will continue to convert existing capacity of traditional copper foil to high-end products, Defu has scheduled to build another 50kt annual capacity in 2027-28 for PCB copper foil focusing on RTF/HVLP, while TGCF has yet to announce additional capacity plans. This may demonstrate Defu's strength in acquiring critical equipment from both external procurement and self-development, per the company. In addition, due to limited capacity of critical tools from international vendors such as Mifune, we expect spillover opportunities for other vendors, and Taijin New Energy as a Chinese supplier with comprehensive equipment portfolio, including surface treatment machine, may be one of major beneficiaries here.

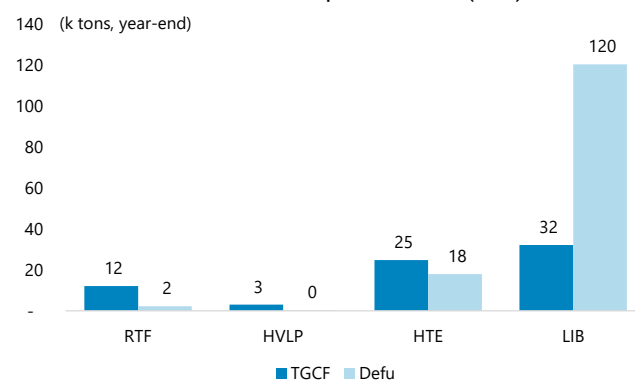
Chart 31 - TGCF vs Defu on Copper Foil Annual Capacity



Source: Company data, Jefferies

Note: Defu's capacity includes 175kt at end-2025 and 20kt from newly acquired Anhui Huiru Technology

Chart 32 - TGCF vs Defu on Product Shipment Structure (2025)



Source: Company data, Jefferies

Implication to our coverage: Sytech to benefit from active capacity expansion from local upstream suppliers. As upstream material supplies, including fiberglass cloth and copper foil, become increasingly tightened, we expect the ability to secure sufficient materials to become another critical differentiator for CCL suppliers in AI supply chain going forward. We also observe a trend that Chinese suppliers are more proactive in expanding production capacity compared to international vendors. Therefore, as a Chinese player, Sytech will likely have a greater advantage in supply chain security going forward, as it continues to diversify its supply sources, enabling it to further expand its market share in AI CCL supply chain.

Company Valuation/Risks

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(Article 3(1)e and Article 7 of MAR)

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			IB Serv./Past12 Mos.		JIL Mkt Serv./Past12 Mos.	
	Count	Percent	Count	Percent	Count	Percent
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